DEEP LEARNING – INTRODUCTION

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1. What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers to learn complex patterns from large amounts of data.

Deep Learning automatically performs feature extraction and representation learning.

It is inspired by the structure of the human brain.

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2. Artificial Neural Network (ANN)

An Artificial Neural Network consists of:

- Input Layer
- Hidden Layers
- Output Layer

Each layer contains neurons (nodes).

Each neuron performs:

1. Weighted sum
2. Add bias
3. Apply activation function

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3. Components of Neural Networks

3.1 Weights

Weights control the importance of input features.

3.2 Bias

Bias helps shift the activation function.

3.3 Activation Functions

Common activation functions:

- ReLU (Rectified Linear Unit)
- Sigmoid
- Tanh
- Softmax

Activation functions introduce non-linearity.

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4. Training Process

Deep learning models are trained using:

1. Forward Propagation
2. Loss Calculation
3. Backpropagation
4. Weight Update

Optimization is done using Gradient Descent.

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5. Loss Functions

For Regression:

- Mean Squared Error (MSE)

For Classification:

- Binary Cross Entropy
- Categorical Cross Entropy

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6. Types of Deep Learning Networks

6.1 Feedforward Neural Network (FNN)

Basic neural network where information flows in one direction.

6.2 Convolutional Neural Network (CNN)

Used for image processing.

Applications:

- Image classification
- Object detection
- Face recognition

6.3 Recurrent Neural Network (RNN)

Used for sequential data.

Applications:

- Language modeling
- Time series forecasting
- Speech recognition

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7. Applications of Deep Learning

- Computer Vision
- Natural Language Processing
- Autonomous Driving
- Medical Diagnosis
- Speech Recognition
- Recommendation Systems

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8. Advantages of Deep Learning

- Automatic feature extraction
- High accuracy
- Works well with large datasets

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9. Limitations

- Requires huge data
- High computational cost
- Needs GPU hardware
- Black box nature

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10. Key Terms

Neural Network
Neuron
Weight
Bias
Activation Function
Backpropagation
Loss Function
Optimizer
Epoch
Batch Size

DEEP LEARNING – ADVANCED CONCEPTS

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1. Backpropagation Algorithm

Backpropagation is used to calculate gradients of the loss function.

Steps:

1. Compute output error
2. Propagate error backward
3. Update weights using gradients

Uses chain rule of calculus.

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2. Optimization Algorithms

2.1 Stochastic Gradient Descent (SGD)

Updates weights per training example.

2.2 Momentum

Speeds up convergence.

2.3 Adam Optimizer

Adaptive learning rate method.
Combines Momentum + RMSProp.

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3. Convolutional Neural Networks (CNN) – Deep Dive

CNN components:

- Convolution Layer
- Filter / Kernel
- Stride
- Padding
- Pooling Layer
- Fully Connected Layer

CNN reduces parameters using weight sharing.

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4. Recurrent Neural Networks (RNN) – Deep Dive

RNN handles sequential dependencies.

Problems:

- Vanishing Gradient
- Exploding Gradient

Advanced RNN variants:

- LSTM (Long Short-Term Memory)

- GRU (Gated Recurrent Unit)

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5. Regularization in Deep Learning

Techniques:

- Dropout
- Batch Normalization
- Early Stopping
- Data Augmentation

Prevents overfitting.

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6. Transfer Learning

Using pre-trained models for new tasks.

Examples:

- ResNet
- VGG
- BERT

Transfer learning reduces training time.

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7. Generative Models

- GAN (Generative Adversarial Network)
- Autoencoders
- Variational Autoencoders (VAE)

Used for:

- Image generation
- Data synthesis
- Style transfer

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8. Hardware Acceleration

Deep Learning training uses:

- GPU
- TPU
- Parallel computing

Large models require high memory and compute.

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9. Model Evaluation in Deep Learning

Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC Curve
- AUC Score

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10. Important Advanced Keywords

Backpropagation
Gradient
Vanishing Gradient
Exploding Gradient
Dropout
Batch Normalization
Transfer Learning
CNN
RNN
LSTM

GAN
Optimizer

🔥 **RAG Optimization Tip**

PDF banate waqt:

- CNN aur RNN ke diagrams add karo
- Neural network ka structure diagram add karo
- Keywords repeat naturally
- 5–7 pages bana do

Isse:

- Better chunking
- Better embedding
- Better retrieval
- Strong hybrid search